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RESEARCH ARTICLE

Benchmarking CNN Architectures for Tool Classification: Evaluating CNN Performance on a Unique Dataset Generated by Novel Image Acquisition System

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ABSTRACT In this study, we introduce the ToolSurface-144 dataset, which is presented here for the first time. It comprises four subsets – Full R, Full S, Top R, and Top S – each containing 144 tool classes captured under varying illumination conditions and fields of view. A newly developed, patented imaging approach was employed to acquire the data. It is compared with conventional diffuse ring illumination to assess its effectiveness in evaluating state-of-the-art convolutional neural networks. This enabled a more targeted investigation of the role of global shape characteristics such as silhouettes versus localized features like the tool face, cutting edges, and delicate geometrical structures under different training strategies. In this study, we evaluate six state-of-the-art convolutional neural networks—AlexNet, DenseNet161, EfficientNet-B0, ResNet152, ResNet50, and VGG16—using three training strategies: fine-tuning, freezing of pre-trained layers, and training from scratch. The results show that EfficientNet-B0 consistently achieved the highest classification accuracy in nearly all experiments and data sets. Especially using the fine-tuning training strategy, the model achieved 99% accuracy in tool classification. ResNet50 benefited greatly from fine-tuning and freezing, achieving a significant increase in performance compared to training from scratch. In contrast, ResNet152, AlexNet, and VGG16 consistently showed poor classification performance, indicating difficulties regarding learning and generalisation. The results show that diffuse illumination and complete tool views provide the best classification conditions, while restricted image sections with homogeneous illumination negatively affect model performance. Among the evaluated training strategies, fine-tuning proved the most efficient training method for developing CNN models for tool classification.

INDEX TERMS Tool classification, machining tools, CNN benchmarking, transfer learning, generalization performance, illumination conditions, multi-class classification, ResNet, EfficientNet-B0, VGG16.

I. INTRODUCTION

Machine vision, in combination with machine learning (ML), has made enormous progress in recent years. In particular, deep learning (DL) has been established in numerous industrial applications and research. Deep neural networks (DNNs) have demonstrated outstanding performance across various applications while exhibiting high scalability, making them applicable to various industrial, commercial, and scientific fields [1].

In computer vision, object classification plays a crucial role. Applications include remote sensing, where DL-based models are widely used for earth observation, detecting, and classifying land use and land cover (LULC) [2].

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Another important application area for DL technologies is medical imaging, which enhances diagnostic accuracy by analyzing X-ray, MRI, and CT images [3], [4]. DL models are also integral to autonomous driving, contributing to the recognition and classification of pedestrians, vehicles, and traffic signs [5], [6]. Furthermore, DL is playing an increasingly important role in agricultural image processing, particularly in monitoring crop conditions and predicting yields using drone images [7], [8].

Several benchmark datasets have recently been established to evaluate and validate DL architectures. A prominent example is ImageNet-1K, which contains over 1.2 million training images spanning more than 1,000 object classes [9]. Other widely used datasets, such as COCO (Common Objects in Context), feature images of natural scenes containing everyday objects and are optimized for object recognition [10]. This broad range of applications highlights the importance of datasets for evaluating DL architectures in object recognition and classification.

However, feature extraction and pattern recognition can be challenging for specific classification tasks. Pixel-based classification approaches, which analyze individual pixels, are often computationally intensive and struggle to capture higher-order image patterns [11], [12]. In contrast, object-based approaches allow extracting meaningful structures within an image but may face difficulties distinguishing visually similar objects [2]. Recent research suggests that scene-based classification [13], which focuses on learning semantic representations of complex image structures, is a promising strategy for improving generalization ability and classification accuracy.

A more specialized yet underexplored area is the classification of complex, symmetric, and reflective objects, such as cutting tools used primarily in metal processing. The application of DL architectures for classifying such tools has been scarcely investigated, with only limited research available on five tool types [14]. One key challenge is the availability of a reliable dataset to facilitate feature extraction and pattern recognition for tool classification, particularly when using pre-trained models originally developed for entirely different datasets. Given that cutting tools' edge structures and reflectance properties differ significantly from everyday objects, conventional DL models may struggle to extract meaningful features.

Our research indicates that little work has been conducted on validating and evaluating DL models for tool classification. One possible reason is the complex geometry of cutting tools and their optical properties—especially their reflective surfaces and intricate surface structures—which pose significant challenges for image acquisition and classification [15]. While numerous studies on wear detection for cutting tools exist, they are predominantly limited to smaller image regions or individual tool types [16].

Recent research has shown that illumination plays a crucial role in image processing and object recognition, particularly in the semantic segmentation of wear patterns. A key question is how illumination, along with the consideration of fine versus coarse features, influences the classification accuracy of neural networks.

This work investigates the performance of state-of-the-art neural networks (SOTA) under different illumination conditions and varying training strategies. The goal is to compare different architectures to identify which strategies are particularly advantageous for the classification of cutting tools. Specifically, the ''training from scratch'' and ''fine-tuning'' approaches are evaluated on pre-trained networks.

This study aims to determine which deep learning architectures are best suited for tool classification and which training strategies yield the highest generalization and classification accuracy under varying lighting conditions and levels of detail.

TABLE 1. List of acronyms.

Acronym	Definition
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
DNN	Deep Neural Network
ML	Machine Learning
MCC	Multi-Class Classification
SOTA	State of the Art
R&D	Research and Development
THI	Technische Hochschule Ingolstadt
RGB	Red Green Blue (Color Space)
LULC	Land Use and Land Cover
MRI	Magnetic Resonance Imaging
CT	Computed Tomography
COCO	Common Objects in Context
ILSVRC	ImageNet Large Scale Visual Recognition Challenge
GAN	Generative Adversarial Network
FLOPs	Floating Point Operations Per Second
MLP	Multilayer Perceptron
ViT	Vision Transformer
MBConv	Mobile Inverted Bottleneck Convolutions
BA-ABC	Bat Algorithm - Artificial Bee Colony
MNIST	Modified National Institute of Standards and Technology dataset
CIFAR	Canadian Institute For Advanced Research

A. ABBREVIATIONS AND ACRONYMS

Table 1 lists the abbreviations and acronyms used throughout this paper for convenience.

II. STATE OF THE ART

A. ARTIFICIAL NEURAL NETWORKS: MILESTONES AND BREAKTHROUGHS

The groundbreaking discovery of McCulloch and Pitts [17] laid the foundation for the development of neural networks. Their research presented the first conceptual model of an artificial neuron based on logical operations, serving as the basis for later developments in this field. Almost 15 years later, in 1958, Frank Rosenblatt developed the first artificial neural network with a learning algorithm: the perceptron [18]. This model represented a significant advance in the research application of artificial intelligence (AI). Rosenblatt's work demonstrated that neural networks could

solve classification problems for simple text and images. Another 28 years later, Rumelhart et al. developed the back-propagation algorithm [19] in 1986, which, for the first time, enabled efficient weight optimization in neural networks. His work ''Learning Representations by Back-Propagating Errors'' laid the basis for training deep neural networks. In 1998, around 10 years later, Yann LeCun et al. introduced the CNN architecture (Convolutional Neural Network)—a milestone in image classification and object recognition [20]. This architecture, known as LeNet-5, paved the way for many later developments. Early practical applications included face recognition and medical image processing. 2012 marked another milestone in developing AI-powered image processing systems: Krizhevsky et al. developed AlexNet. This deep CNN architecture won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) by a large margin [21]. This success was primarily due to the use of GPUs, enabling the practical training and optimization of deep neural networks. Research continued at a rapid pace. Goodfellow et al. introduced the Generative Adversarial Network (GAN) algorithm in 2014, significantly influencing image generation and AI-driven creative applications [22]. At the same time, the VGG architecture (Visual Geometry Group) was introduced, a deep CNN architecture with 16 layers that used simple 3×3 convolutional filters [23]. VGG-16 achieved an accuracy of 71.5% on the ILSVRC dataset but had a high computational cost. However, as neural networks grew deeper, challenges such as vanishing gradients emerged, making training more difficult. This led to the introduction of the ResNet (Residual Networks) architecture by He et al. in 2015 [24]. The skip connections in ResNet enabled a better gradient flow, making very deep networks trainable. The ResNet-50 and ResNet-152 variants achieved accuracy on the ILSVRC dataset of 76.6% and 78.3%, respectively. Further advancement came with DenseNet-161, developed by Huang et al. [25]. This architecture introduced the concept of dense connections, where each layer is connected to all previous layers, allowing it to require fewer filters per layer. DenseNet-121 required only 8M parameters, while ResNet-50 had around 25M and ResNet-152 around 60M parameters. In addition to reducing training parameters, DenseNet-121 showed better generalization on smaller datasets. Another significant advance was made in 2019 with the introduction of the EfficientNet architecture by Tan and Le [26]. This architecture used optimized scaling of depth, width, and resolution and introduced the MBConv (Mobile Inverted Bottleneck Convolutions) blocks from MobileNetV2. As a result, EfficientNet required less computing power than previous CNNs. Various variants were developed, ranging from EfficientNet-B0 (5.3M parameters) to EfficientNet-B7 (66M parameters). On the ImageNet dataset, EfficientNet-B7 achieved an accuracy of 84.3%, while EfficientNet-B0 achieved 77.1%. In 2021, the research team led by Li et al. presented an extended version called EfficientNet-X. This architecture achieved 86% accuracy on ImageNet with an even more efficient computing strategy—using 30% fewer FLOPs than EfficientNet-B7. While CNNs continued to perform, a new approach to image processing was introduced in 2020: Vision Transformers (ViT) [27]. This architecture, developed by Dosovitskiy et al., transferred the Transformer model from NLP research (2017) to images. ViT replaced classical CNNs with a self-attention mechanism, achieving an accuracy of 85.2% on ImageNet and outperforming EfficientNet-B7 (84.3%). However, ViT required significantly more computing power at 191B FLOPs compared to EfficientNet-B7 (37B FLOPs). A novel approach emerged in 2021: MLP-Mixer, developed by Tolstikhin et al. [28], which replaced Convolutional Layers and Self-Attention with pure MLPs (Multilayer Perceptrons) for the first time. This architecture showed potential for real-time and edge computing applications. Another significant development came in 2023 when Mingxing Tan et al. presented the hybrid CoAtNet architecture [29]. This combines the strengths of CNNs and Transformers, where early layers rely on convolutions while later layers use self-attention. CoAtNet-7, applied to ImageNet-1K, achieved a Top-1 accuracy of over 87.9%—one of the highest recorded to date.

B. AVAILABLE DATASETS FOR THE VALIDATION OF NEURAL NETWORKS

This section presents the most relevant datasets for training and evaluating neural networks. We start with the **MNIST dataset**, published in 1998. It contains 60,000 training images and 10,000 test images of handwritten digits (0-9), each with a resolution of 28×28 pixels. MNIST is considered one of the first and most widely used base datasets for simple classification tasks [20].

As research progressed, larger and more diverse image datasets were developed. **CIFAR-10 and CIFAR-100**, published in 2009, are among the first comprehensive datasets for general image classification tasks. CIFAR-10 includes 60,000 color images of size 32×32 pixels divided into 10 classes, including vehicles such as cars and natural objects such as animals. CIFAR-100 is an extended version with 100 more finely subdivided classes and is often used as a benchmark for smaller image datasets [30].

The need for larger and more complex datasets grew alongside advancements in artificial intelligence and increasingly powerful neural networks. **ImageNet**, published in 2010, is one of the most essential image datasets for training deep neural networks, containing over 1,000 different classes. It played a central role in developing powerful models such as AlexNet and ResNet, used in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) [9].

Alternative datasets were developed to accommodate specific use cases. **Fashion-MNIST**, an extension of MNIST, was introduced as an alternative benchmark for the image classification of fashion items. The dataset contains 70,000 images with a resolution of 28×28 pixels and covers 10 categories, including shoes, pants, and jackets [31].

Another notable addition is **Tiny ImageNet**, introduced in 2017 as a reduced version of ImageNet. It contains 100,000 images with a resolution of 64 × 64 pixels and is particularly useful for evaluating models on smaller datasets [32].

The evaluation and benchmarking of SOTA CNN networks is usually performed using either model-centric or data-centric approaches. In the work of Bilal et al., for example, a hybrid approach was presented in which a separate CNN (XcelNet17) is used in combination with a metaheuristic algorithm (BA-ABC) for feature selection. The architecture was validated with networks such as AlexNet, VGG16, ResNet50 and DarkNet19 on general remote sensing datasets such as AID with 30 classes on RGB images [33]. As an example of a data-centric approach, the work of Hafiz et al. used pre-trained models such as AlexNet, VGG16, VGG19, ResNet50, InceptionV3, DenseNet121 and different variants of EfficientNet. The underlying dataset consisted of 3264 MRI images of different tumour classes (including 926 glioma images and meningiomas), supplemented by data augmentation methods and colour noise suppression using FastNlMeans denoising [34].

In this work, four unique datasets composed of different, complex cutting tools are used to validate SOTA classification algorithms and gain new insights into the training capabilities of various neural networks. These tools, comprising 144 different classes, are unique due to their complex shapes and optical properties, including reflection and scattering on the surface. The cutting tools include standard tools such as milling, drilling, reaming, tapping, and various specialized tools.

III. METHODOLOGY

The performance of neural networks depends not only on the choice of architecture and classification approach but also on image quality and the features extracted from the images. In the case of cutting tools, the detection system, particularly the lighting and sensor technology, plays a central role in highlighting relevant features. As described in Section II, previous datasets have entirely different visual characteristics, primarily consisting of images of people, animals, plants, or everyday objects. To the best of our knowledge, a comparable dataset focused on cutting tools does not exist, making the investigation of existing models' performance on this novel dataset particularly valuable for the computer vision community.

The primary task of the optical illumination system is to generate high-quality, stable, and reproducible images [35], [36]. The lighting must be designed to enhance relevant wear features while minimizing unwanted reflections and shadows. However, capturing high-quality images of metallic and complex-shaped surfaces, such as those found on cutting tools, presents a considerable challenge [37], [38]. The coatings on cutting tools further compound this challenge, as they can degrade image quality or even render precise inspection impossible. Therefore, specialized lighting techniques are required to enable effective inspection.

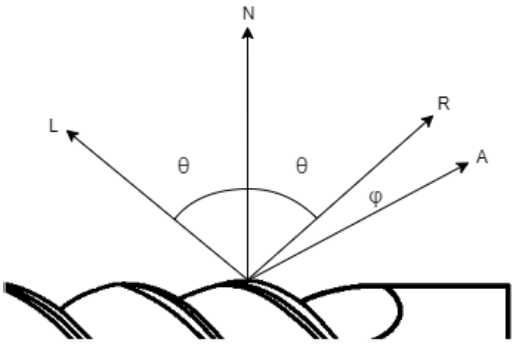


FIGURE 1. The Phong model is utilized to simulate the reflection behaviour of light. The reflective component "R" results in a specular light that depends on the orientation of the surface of the tool concerning the observer (A), the normal vector (N), the point light source (L) and R as a unit vector in the direction of the ideal specular reflection. θ is the viewing angle relative to the specular reflective direction R, and ϕ finally represents the angle L and R form to N. Consequently, complex milling tools can have varying gloss, shadowing or reflectivity areas, which are influenced by factors such as the tool material, shape or surface finish.

In industrial image processing for wear detection, such as on milling tools, circular LED rings [16], diffuse direct lighting [39], or dome lighting [40] are widely used. These light sources offer the advantage of high-intensity illumination and can be easily installed on various devices. They help reduce shadows and enhance the visibility of relevant features for analysis. Despite the benefits of diffuse lighting, which is particularly useful for inspecting reflective surfaces, it cannot entirely eliminate unwanted reflections.

Recent advancements in lighting methods have demonstrated that illumination is crucial in improving inspection accuracy [15], [41], [42], [43], [44]. The novel illumination units introduced in [15] have never been used for classification tasks.

A. LIGHT BEHAVIOUR ON THE SURFACES OF CUTTING TOOLS

Light behaviour on the surfaces of cutting tools: Unwanted light artefacts are usually caused by different curvatures and orientations of the tool surface, which lead to regular and specular reflections. As shown in Figure 1, end mills have various curved surfaces with reflective properties. Even with a direct, diffuse light source, avoiding reflections to an acceptable degree remains challenging.

The Phong reflection model describes the composition of the perceived brightness of an object's detected light intensity (I). Three components should be taken into account: Ambient, diffuse and specular Light [45]. The detected intensity I is given as follows:

$$I = I_a K_a + I_l K_a \cos\theta + I_l K_d \cos\theta + I_l K_s \cos\alpha \quad (1)$$

where K_a, K_d and K_s are the ambient, diffuse, and specular light coefficients, respectively. I_a and I_l are the intensities of ambient and direct light from the source, respectively. θ is the angle between the incident light and the standard line of the

object surface. α is the angle between the reflected light and the line of sight, and n is the specular reflection coefficient.

The ambient component arises from the reflection of omnidirectional light from an infinitely large source. The diffuse component considers the scattering of light on rough surfaces and is reflected randomly in all directions. The specular component is based on perfect specular reflection, as typically occurs on metal surfaces. Since we focus on characterising metallic surfaces of an end mill with high optical quality and lower surface roughness, the diffuse component can be neglected, and only the reflective and ambient light components should be considered. According to [46], the perceived brightness is then given by

$$I = K_b I_b + K_d I_d \cos\theta + K_r I_r \tag{2}$$

where K_a, K_d and K_s are the coefficients of ambient, diffuse, and specular light, respectively. I_a and I_l are the intensities of ambient and direct light from the source, respectively. θ is the angle between the incident light and the normal line of the object surface. α is the angle between the reflected light and the line of sight, and n is the specular reflection coefficient. For the first time, we utilise the newly patented measurement setup [47], shown in Figure 2a, previously used for CNN-based wear detection to benchmark CNN architectures. Thus, as part of this work, a new type of optical detection system was used for the first time to classify cutting tools. The system enables homogeneous illumination of cylindrically symmetrical components, such as end mills, to ensure precise, image-based quality control. Even the smallest effects and different signs of wear can be reliably imaged. The special feature lies in combining a structurally optimized housing shape, multiple diffuse light guidance and a specially arranged LED emitter unit. The system is supplemented by specifically aligned baffles and a specially selected coating to promote multiple light scattering.

The optical housing forms the central component of the system. It is designed to form a closed inner chamber and consists of a cylindrical hollow body and a dome-shaped semi-dome (2a (4) and 2a (2)) with a special diffuser and LED array (2a (3)). The object to be examined is located exactly in the centre of the closed enclosure (2a (1)). The outer space of the housing also serves to shield the object from ambient light. The inner walls are coated with a lightscattering, highly reflective material - typically barium sulfate. The dome contains a ring-shaped, radially symmetrical diffuser (2a (3)). This is designed so that its concave surface (2b) faces the top of the dome-shaped housing (2a (3)), and incident electromagnetic radiation is diffusely scattered in the direction of the inner walls.

The scattered light is guided in several successive phases:

1) **first scattering** at the concave diffuser surface (2b), which is coated with barium sulphate, for example;
2) **second scattering** at the inner dome wall (2a(3);
3) **additional scattering** due to reflections on other inner surfaces ((2a (4)) or on specifically positioned baffles (not shown), which also consist of scattering material.

For optimum illumination of the tools, 14 LEDs were used as emitters, which are arranged in a ring and radially around the axis of symmetry (2a (3)) and aligned with the diffuser (2b). The radiation reflected back from the object is guided through an optical aperture to an image acquisition system, e.g. a high-resolution camera unit with a lens.

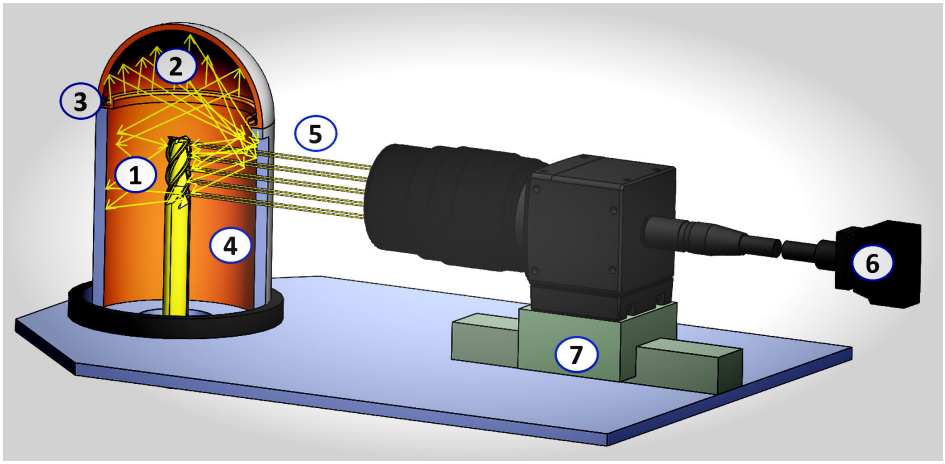
The camera is mounted on a holder that can be used to adjust the distance to the object using a motor (2a (6) and (7). The actuator is a rotation plate with a three-jaw chuck mounted centrally within the inner housing. This allows the tool to be fixed and rotated in steps. In the version realized here, images were generated in steps of **7.5**°, which enables a complete inspection of the tool surface over **360**°. As a comparison, Figure 2c shows the same measurement setup with a classic diffuse ring light, which has previously been used to inspect metallic components. These two variants are used to generate data sets when evaluating the SOTA-CNNs. Figure 3 clearly illustrates the difference between images captured using the patented acquisition system shown in Figure 2a and those taken with the system from Figure 2c. It is evident that images captured with direct, diffuse ring lighting exhibit strong reflections on the curved surfaces, increasing towards the center and accompanied by unfavourable color representation. In contrast, images taken with the new patented acquisition system effectively suppress or nearly eliminate such reflections, resulting in a more homogeneous, low-reflection image with improved color fidelity and depth of field.

B. NEURAL NETWORK ARCHITECTURES

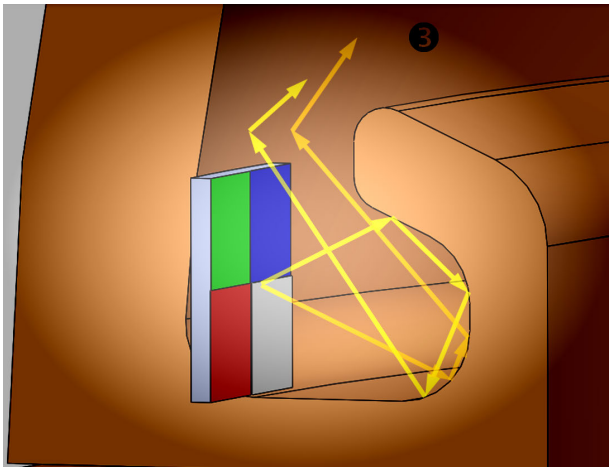
The use of deep learning techniques to automatically classify tools under different lighting conditions opens up new perspectives for evaluating neural networks. Due to the tool's complex geometric properties and symmetry, this data set is unique for such investigations. Evaluating different model architectures using this dataset is essential to providing a solid basis for further research.

This study focuses on the analysis of certain architectures:

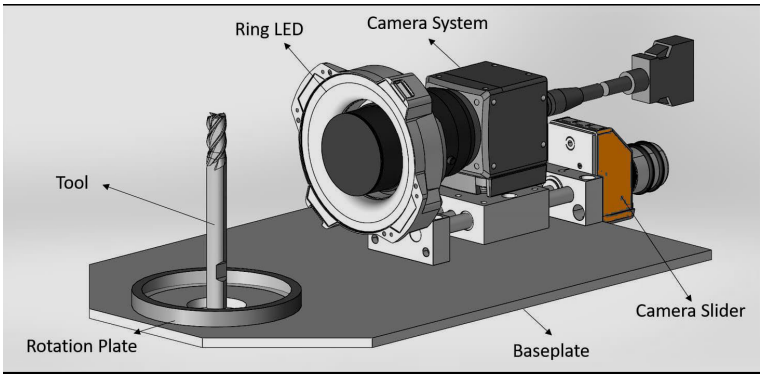
- AlexNet [48], a pioneering model, has significantly advanced the use of deep neural networks in image recognition. Despite its relatively simple architecture, it offers valuable insights into the fundamentals of image classification.
- VGG16 [49], known for its deep structures that enable the capture of complex patterns and whose uniform architecture provides deep insights into convolutional networks.
- ResNet50 [50] and ResNet152 [51], which remain powerful even with increasing depth of the networks, thanks to their innovative residual blocks, thus providing an excellent basis for comparison.
- DenseNet161 [52], whose dense connection structure allows for effective learning of features at different levels of abstraction.



(a)



(b)



(c)

FIGURE 2. **(a): The housing consists of a cylindrical cavity (1) with a hemisphere (2) on the upper side. Thirteen LEDs are arranged as spotlights at regular intervals along the U-shaped edge of the hemisphere (3). A tool is located in the centre of the housing (4). The output opening leads to the camera (5). A Serial interface connects the camera to the computer (6). The camera is fixed or mounted on a camera mounting plate. (b) An illustration of the U-shaped internal reflector, which improves the multiple scattering of light. (c): Experimental setup for tool imaging with a rotary table for positioning the tool, a ring LED for direct, diffuse illumination, a camera system mounted on a slider for precise settings and a stable base plate for system stability and reproducibility.**

- EfficientNet-B0 [53] promises high performance with comparatively low computing effort due to its scaling efficiency and thus seems particularly suitable for resource-conscious applications.



(a) (b)

FIGURE 3. A comparison of images acquired with different lighting methods: (a) directdiffuse lighting and (b) novel light source.

Considering their widely used research work and ability to process complex image data, these architectures are compared. The evaluation criteria include classic performance indicators such as accuracy, precision, recognition value and F1 score, resistance to overfitting, and the ability to generalize to unknown data. The data set should also enable other scientists to use it to validate other architectures and methods in the future.

C. RESOURCES AND COMPUTATIONAL SETUP

All experiments were conducted using a high-performance computing cluster at AImotion Bavaria. The following resource allocations were utilized for model training:

- Maximum walltime per job: 5 days (120:00:00 hours).
- Up to 40 CPU cores and 300 GB RAM per job.
- GPU allocation: One NVIDIA A100 80GB PCIe GPU partitioned to 40GB of usable memory.

The models were implemented using the PyTorch framework (version 2.0.0) on a Linux-based system. CUDA 11.8 and cuDNN were employed for GPU acceleration, and Python 3.10 was used for scripting. Random seeds were fixed to ensure reproducibility across training runs.

D. DATASET

In this work, we evaluate the performance of several established CNN architectures using standard metrics such as precision, recall and F1-score. Current architectures such as AlexNet, DenseNet161, EfficientNet-B0, ResNet152, ResNet50 and VGG16 are used for evaluation. Our analysis focuses on EfficientNet-B0 as the only representative of the EfficientNet model series. We follow a data-centric approach here with unique tool images captured under industrial conditions and under different lighting scenarios. We consider two fields of view: one of the entire tool and one of the upper part only. The images were captured using classic diffuse ring lighting and a novel, patented imaging system that provides high-quality images [47]. The data sets contain different tools with different geometries and illumination conditions of 144 tool classes. Concretely, the four data sets differ in this respect:

- Dataset Full S and Top S (Figure 6a and Figure 6b): Focus on the entire tool and the upper part of the tool under uniform illumination to enable comprehensive detection and detailed inspection.
- Dataset Full R and Top R (Figure 7a and Figure 7b): Contain images of the entire tool and its upper part under direct diffuse ring illumination, enhancing contouring and detail recognition.

As shown in Table 2 and Figure 4 the image size of the four datasets varies due to the different sizes and shapes of the tools, if only the entire tool is considered, then after scaling to 224 × 224 pixels fine features are suppressed in the entire image of the tool, while considering only the upper area of the tools delicate structures such as roughness or various technical parameters, including those shown in Figure 5 are more apparent when extracting features.

The tools include drills, end mills, chamfers, flat grinders, multi-drills, step drills, twist drills, and countersinks.

To highlight the specificity of the dataset, some of the geometric and technical parameters of four different tools are shown in Figure 5. These features can play a crucial role in training neural networks. Since the images of the tools are scaled to 224 × 224 pixels, finer details may be lost. The differences in the tool images due to the different exposure conditions and tool geometry can be seen in Figures 6 and 7.

For this reason, different datasets are created with varying resolutions of the geometric and technical parameters to ensure that delicate structures, which are only partially visible when capturing the entire tool, are preserved. This is particularly important when downscaling the images. On the other hand, by analyzing the contours more extensively, the neural network may increasingly identify radially symmetrical features as relevant.

E. TRAINING AND EVALUATION PROTOCOL

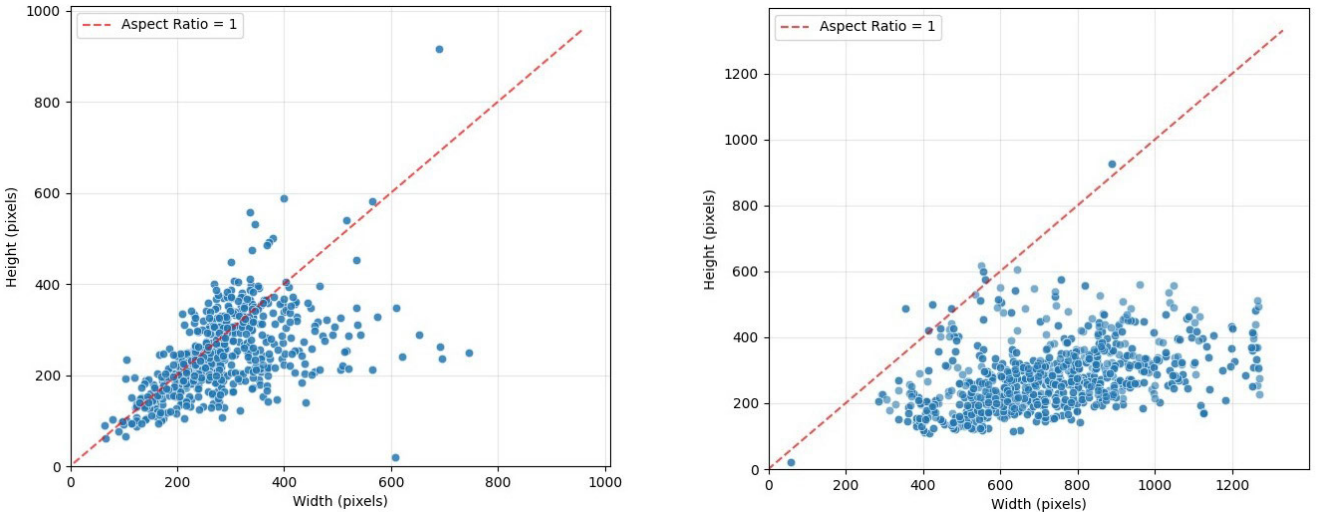
Two datasets were generated under different lighting conditions for tool classification: ring illumination and homogeneous illumination. One dataset contains images of the entire tool, while the other focuses on the upper part, capturing important geometric parameters such as axis, spiral, and tool face angles. This resulted in four datasets: two for each lighting condition (entire tool and upper part).

To ensure reproducibility, all datasets were split into training (60%), validation (20%), and test (20%) sets using stratified random sampling. The models were trained using the same training data, with parameter selection based on the validation set. Early stopping was applied, and the best-performing model (lowest validation loss) was evaluated on the test dataset.

Image preprocessing included resizing to 256 × 256 pixels and cropping to 224 × 224 pixels. Only RGB images were used. Hyperparameter optimization was performed for

TABLE 2. Overview of the four datasets from ToolSurface-144 used to evaluate the CNN models in this study. Each subset contains 144 tool classes with 48 images per class, captured under specific field-of-view and illumination conditions. Due to variations in tool geometry, the original image sizes differ; all images were scaled to a uniform resolution of 224 × 224 pixels before being fed into the CNNS.

Dataset	Classes	Images/class	Color Space	Illumination & Focus	Min Size (px)	Max Size (px)
Full R	144	48	RGB	Entire tool under direct diffuse ring illumination containing high reflection portions from the cutting edge surface.	Width: 404, Height: 112	Width: 889, Height: 927
Full S	144	48	RGB	Entire tool under uniform illumination without reflection and increased depth of field.	Width: 404, Height: 112	Width: 889, Height: 927
Top R	144	48	RGB	Upper part of the tool under direct diffuse ring illumination, highlighting cutting edge details by reflection pattern.	Width: 67, Height: 61	Width: 689, Height: 916
Top S	144	48	RGB	Upper part of the tool under uniform illumination for detailed, reflection-free imaging of cutting parameters.	Width: 67, Height: 61	Width: 689, Height: 916



(a) Top R and Top S subsets

(b) Full R and Full S subsets

FIGURE 4. Scatter plots showing the width and height distributions of images from the Top R and Top S subsets (a) and the Full R and Full S subsets (b) of the ToolSurface-144 dataset. The Top datasets focus exclusively on the upper part of the tools. The red dashed line represents an aspect ratio of 1:1. The observed variation in image dimensions reflects the diversity of tool geometries. It highlights the relationship between the tool shaft and the cutting area.

learning rates of 0.01, 0.001, and 0.0001. The ReduceLROnPlateau scheduler lowered the learning rate when no loss improvement was observed, with a patience factor of 5 and a reduction factor of 0.1. The maximum number of training epochs was set to 100, with early stopping triggered after 10 epochs with no improvement in validation loss. The batch size was set to 128, and the Adam optimizer was used.

To validate the neural networks, all models were trained using the following training strategies:

- **From Scratch**: Training the model from scratch. The weights are initialized randomly. It requires a large amount of data for good results and is associated with a long training time. The method is particularly suitable for data sets similar to ours.
- **Freezing Pretrained Weights**: This method can train the pre-trained models based on a similar dataset. The previous layers of the model are frozen so that the model can benefit from the general features, while the upper layers are adapted to the specific classes. The primary benefit is that training time is reduced and performance is improved.
- **Fine-tuning Pretrained Weights**: This method is particularly suitable when dealing with a small data set with many classes. The pre-trained weights are used as a baseline to better capture the differences and subtleties of the 144 tool classes. It balances efficiency and adaptability and provides a flexible and versatile approach to training.

F. CLASSIFICATION METRICS

To evaluate the performance of the models, the following metrics have been used:

- **Accuracy**: Accuracy measures the proportion of correctly classified instances (positive and negative) out of the total cases. The formula is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

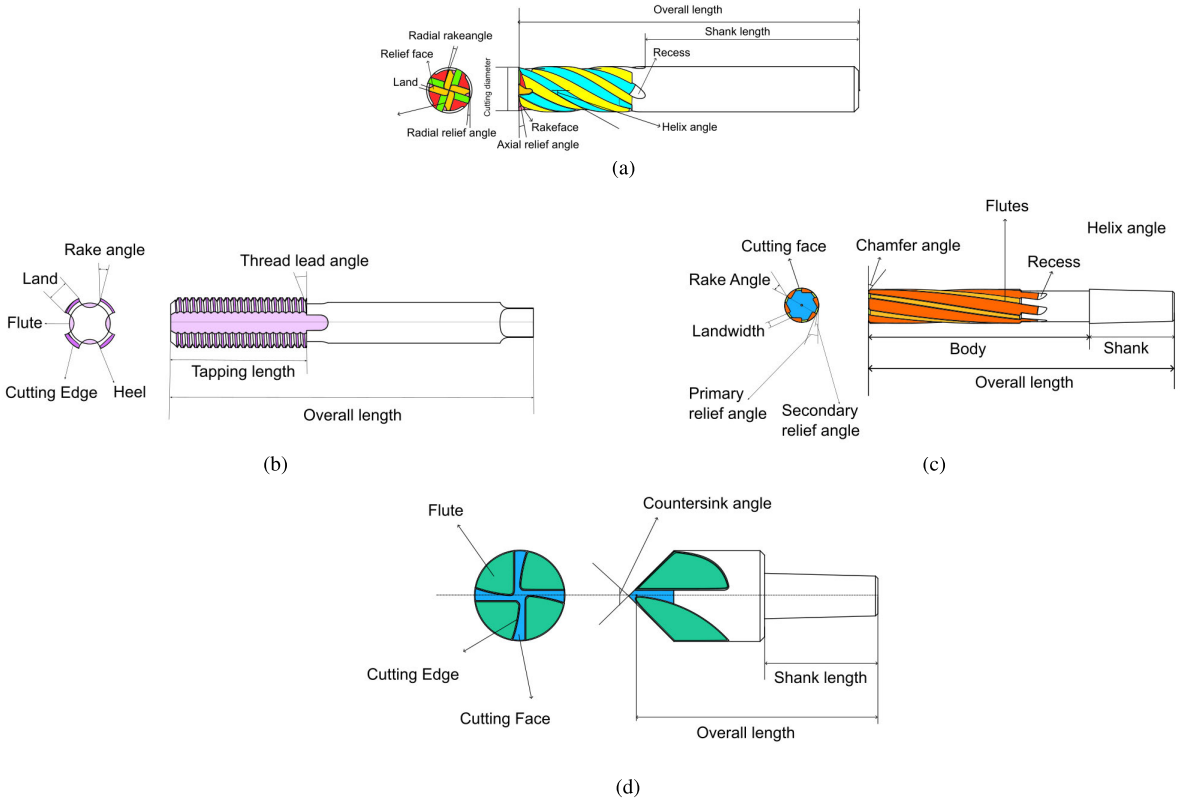


FIGURE 5. **Illustration of the geometric parameters and angles of cutting tools such as:(a) Endmill: radial and axial clearance angle, flute, end face and helix angle.(b) Taps: including thread lead angle, clearance angle, cutting edge and thread length.(c) Reamer: flutes, main and secondary clearance angles, end face and the length of the shank.(d) Countersink: countersink angle, flutes and cutting surfaces.**

- **Precision**: Precision indicates how many of the instances classified as positive are positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision means that the classifier produces few false positives. This metric is critical when the cost of false positives is high.

- **Recall (Sensitivity)**: Recall measures how many of the actual positive instances were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall means that the classifier produces few false negatives. This metric is essential when missing positive instances (false negatives) is costly.

- **F1-Score**: The F1-score is the harmonic mean of precision and recall (sensitivity), balancing both metrics:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP represents True Positives, TN stands for True Negatives, FP for False Positives and FN for False Negatives. Accuracy is helpful if the class distribution is balanced and the costs for incorrect classifications are the same for both classes. A high recall is necessary if the absence of positive instances (false negatives) is costly. High precision is crucial when the cost of false positives is high. The F1 score is helpful when a balance between precision and recall is necessary and the class distribution is unbalanced.

IV. RESULTS AND DISCUSSION

We present the results validating different deep learning (DL) models for multi-class classification (MCC) using four different datasets, each consisting of 144 highly complex, optically critical tools.

Models from six architectures are evaluated: AlexNet, VGG16, ResNet50, ResNet152, DenseNet162, and EfficientNet-B0. For each model architecture, we evaluated three different training strategies: (i) models trained from scratch, (ii) fine-tuned models previously trained on the ImageNet-1K dataset, and (iii) models with frozen training parameters using pre-trained weights.

The following research questions will be discussed:

- Comparison of the performance of different CNN architectures and their generalization capabilities, identifying which models fail during training.
- Investigation of the classification performance of different datasets and answering the research question: Do images captured under diffuse ring light or high-quality

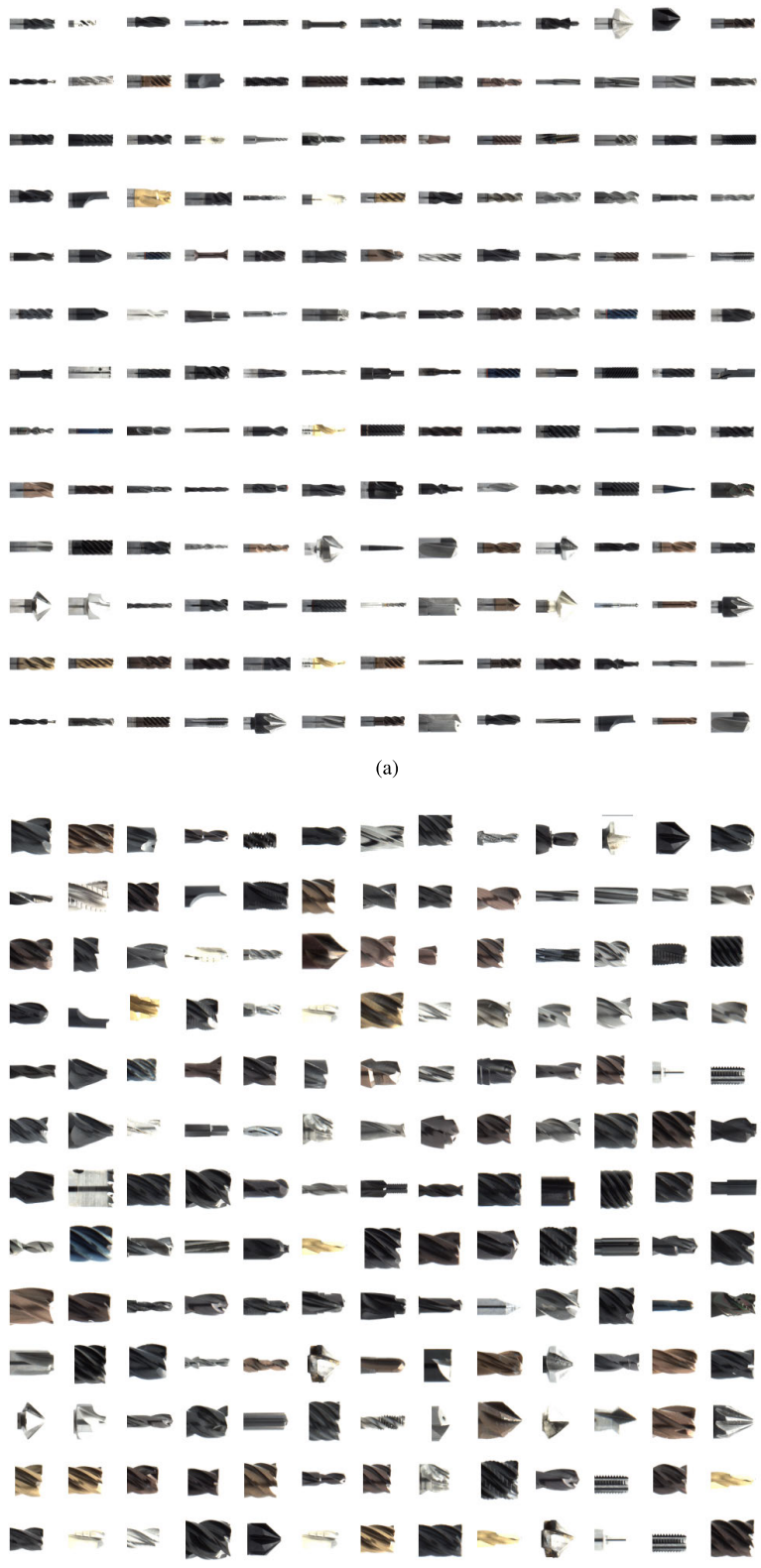
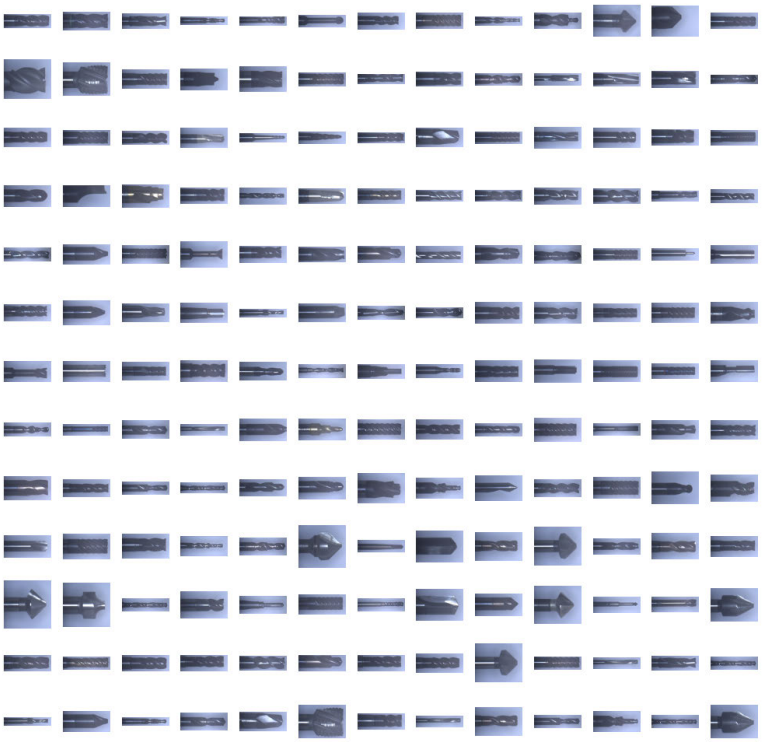
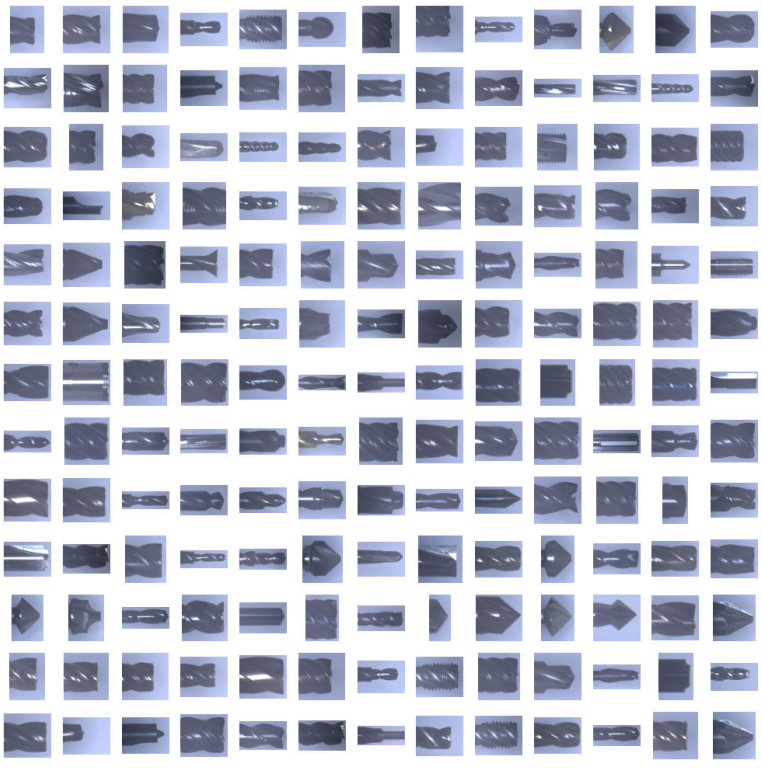


FIGURE 6. Illustration of the data sets of different cutting tools. The data sets were captured under different acquisition conditions. The images of the complete tool (a) and the upper tool segments (b), named Full S and Top S.



(a)



(b)

FIGURE 7. In contrast to Figure 6, the data sets (c) with complete tools and (d) with the upper tool segments originate from the acquisition system (see Figure 2c), named Full R and Top R.

TABLE 3. Accuracy results across models and datasets.

Dataset	Experiment	AlexNet	DenseNet161	EfficientNet0	ResNet152	ResNet50	VGG16
full R	finetuning	0.2328	0.7421	0.9910	0.2634	0.8591	0.2328
	freezing	0.8574	0.6827	0.8623	0.3954	0.4179	0.8175
	scratch	0.2328	0.9588	0.9755	–	0.3167	0.2328
full S	finetuning	0.2448	0.4959	0.9837	0.2567	0.7895	0.2448
	freezing	0.9319	0.6036	0.9348	0.4402	0.3759	0.8502
	scratch	0.2448	0.2917	0.9375	0.2448	0.2742	0.2448
top R	finetuning	0.2223	0.9167	0.9856	0.5517	0.9391	0.2223
	freezing	0.8366	0.3589	–	0.4287	0.3033	0.7574
	scratch	0.2223	–	0.9239	–	–	0.2223
top S	finetuning	0.2251	–	0.9543	0.3399	0.8143	0.2251
	freezing	0.9213	0.6242	0.8676	0.5560	0.4557	0.8434
	scratch	0.2251	0.2251	0.8889	–	–	0.2251

TABLE 4. Precision results across models and datasets.

Dataset	Experiment	AlexNet	DenseNet161	EfficientNet0	ResNet152	ResNet50	VGG16
full R	finetuning	0.0016	0.6449	0.9854	0.0139	0.8728	0.0016
	freezing	0.8369	0.6385	0.8638	0.1831	0.1914	0.7999
	scratch	0.0016	0.9609	0.9782	–	0.1159	0.0016
full S	finetuning	0.0017	0.3846	0.9690	0.0231	0.8147	0.0017
	freezing	0.9096	0.4510	0.9296	0.2032	0.0863	0.8371
	scratch	0.0017	0.0248	0.9301	0.0017	0.0128	0.0017
top R	finetuning	0.0015	0.9037	0.9845	0.4464	0.9167	0.0015
	freezing	0.8307	0.0783	–	0.2244	0.0454	0.7564
	scratch	0.0015	–	0.9087	–	–	0.0015
top S	finetuning	0.0015	–	0.9596	0.2339	0.8627	0.0015
	freezing	0.9029	0.5834	0.8598	0.4132	0.2393	0.8109
	scratch	0.0015	0.0015	0.8871	–	–	0.0015

TABLE 5. Recall results across models and datasets.

Dataset	Experiment	AlexNet	DenseNet161	EfficientNet0	ResNet152	ResNet50	VGG16
full R	finetuning	0.0068	0.6242	0.9899	0.0201	0.8194	0.0068
	freezing	0.8431	0.5674	0.8080	0.1387	0.1824	0.7385
	scratch	0.0068	0.9421	0.9592	–	0.1228	0.0068
full S	finetuning	0.0068	0.3563	0.9765	0.0300	0.7609	0.0068
	freezing	0.8993	0.4072	0.9082	0.2054	0.0906	0.8100
	scratch	0.0068	0.0367	0.9234	0.0068	0.0143	0.0068
top R	finetuning	0.0068	0.8692	0.9836	0.4018	0.9170	0.0068
	freezing	0.7748	0.0901	–	0.2053	0.0550	0.6753
	scratch	0.0068	–	0.9050	–	–	0.0068
top S	finetuning	0.0068	–	0.9637	0.1859	0.7962	0.0068
	freezing	0.8750	0.5013	0.8337	0.3522	0.2076	0.7707
	scratch	0.0068	0.0068	0.8564	–	–	0.0068

TABLE 6. F1-Score results across models and datasets.

Dataset	Experiment	AlexNet	DenseNet161	EfficientNet0	ResNet152	ResNet50	VGG16
full R	finetuning	0.0026	0.6039	0.9851	0.0145	0.8162	0.0026
	freezing	0.8135	0.5518	0.8055	0.1238	0.1520	0.7345
	scratch	0.0026	0.9461	0.9625	–	0.0921	0.0026
full S	finetuning	0.0027	0.3131	0.9698	0.0204	0.7503	0.0027
	freezing	0.8943	0.3759	0.9082	0.1735	0.0695	0.7941
	scratch	0.0027	0.0246	0.9182	0.0027	0.0082	0.0027
top R	finetuning	0.0025	0.8693	0.9829	0.3759	0.9064	0.0025
	freezing	0.7715	0.0694	–	0.1788	0.0330	0.6725
	scratch	0.0025	–	0.8927	–	–	0.0025
top S	finetuning	0.0025	–	0.9573	0.1638	0.7937	0.0025
	freezing	0.8717	0.4805	0.8246	0.3349	0.1824	0.7671
	scratch	0.0025	0.0025	0.8486	–	–	0.0025

images from the patented measurement system yield better results?

- Confirmation of the benefits of transfer learning observed in previous studies on this unique dataset.
- Examination of the data-centric approach and its impact on CNN model performance, addressing the question: Are coarse or detailed images more suitable for tool classification?
- Evaluation of the choice of the pre-trained dataset and its impact on performance compared to training from scratch.

A. TRAINING MODELS FROM SCRATCH

Here, we begin by analyzing the performance of models trained from scratch, i.e., models initialized with random weights during training. Tables 3 - 6 present these results for the MCC. Table 3 shows the accuracy (%) of the models trained from scratch for the four MCC datasets. It also ranks the models based on their performance across the four datasets.

Table 3 indicates that EfficientNet-B0 stands out among the convolutional architectures. EfficientNet-B0 achieves the highest accuracy across all datasets (full R: 0.9755, full S:

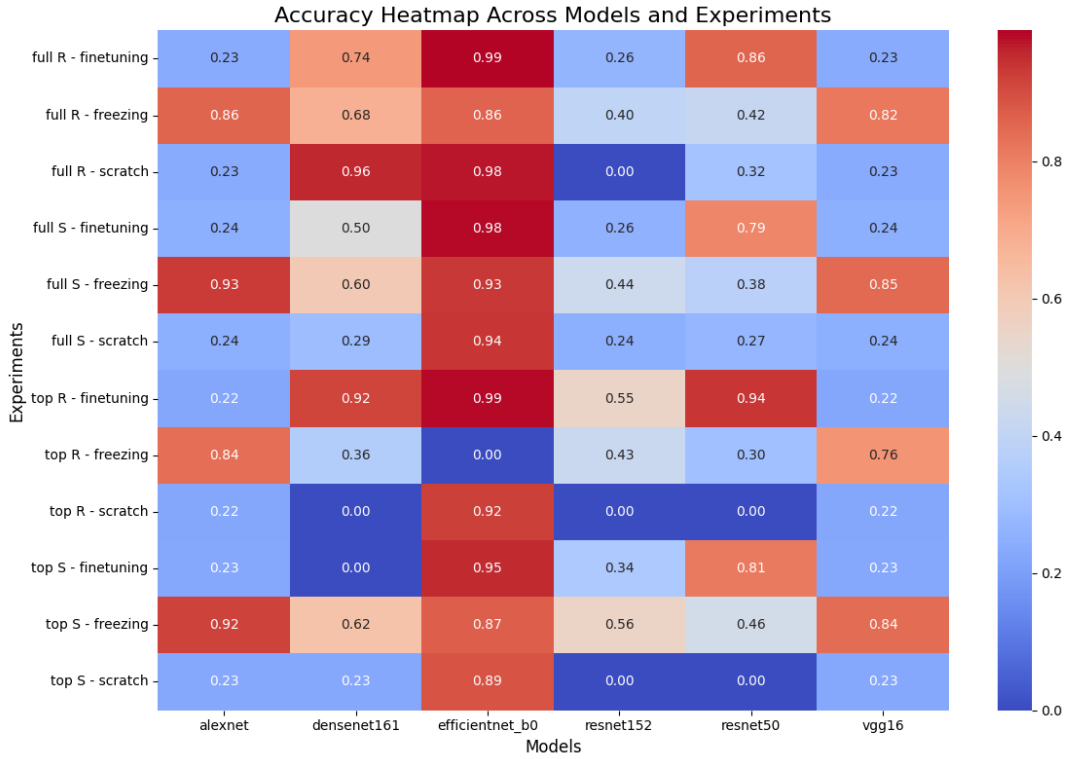


FIGURE 8. Accuracy heatmap across models and datasets.

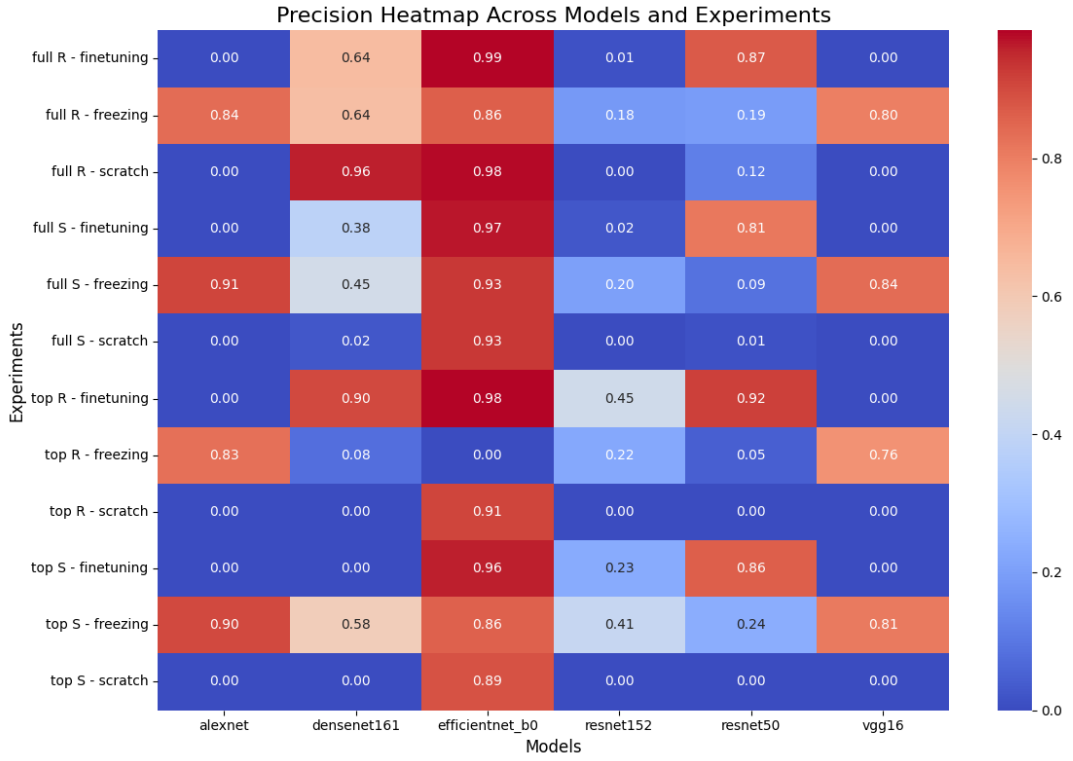


FIGURE 9. Precision heatmap across models and datasets.

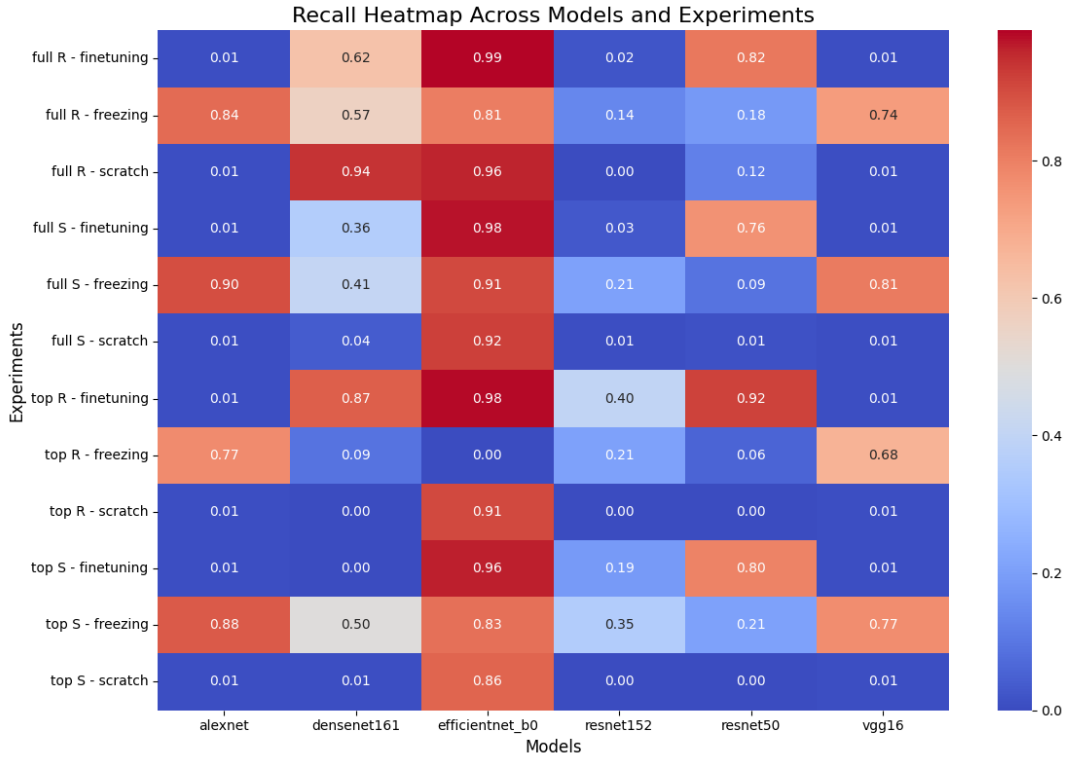


FIGURE 10. Recall heatmap across models and datasets.

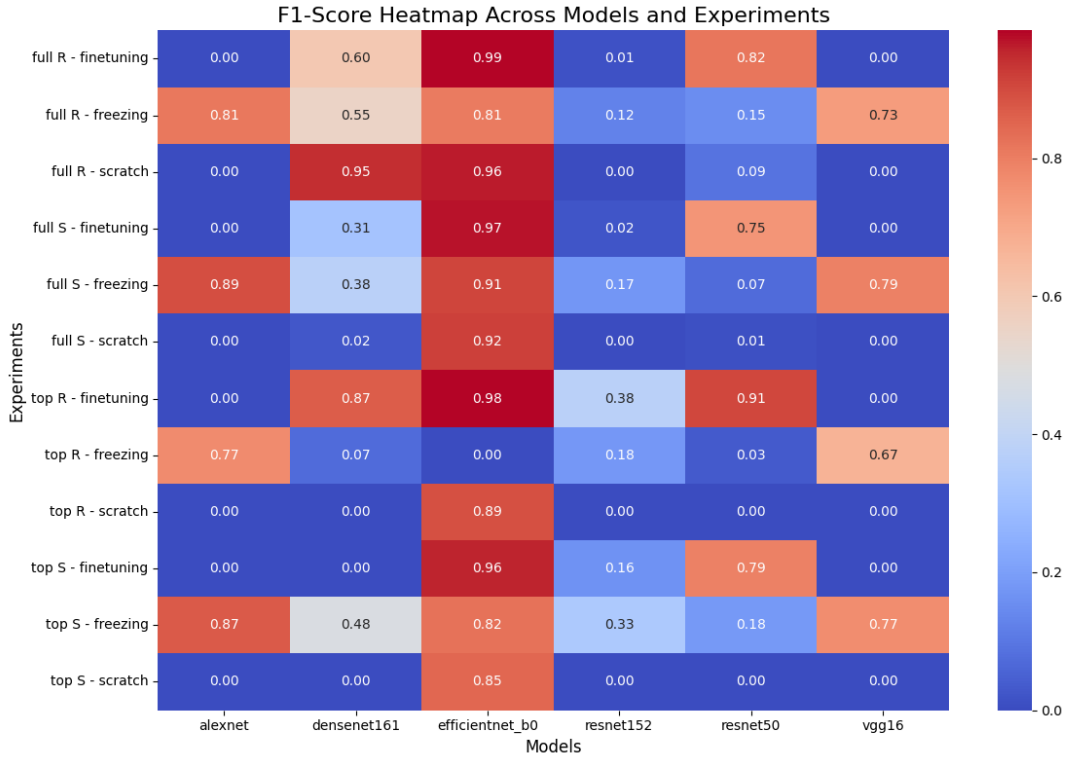


FIGURE 11. F1-Score heatmap across models and datasets.

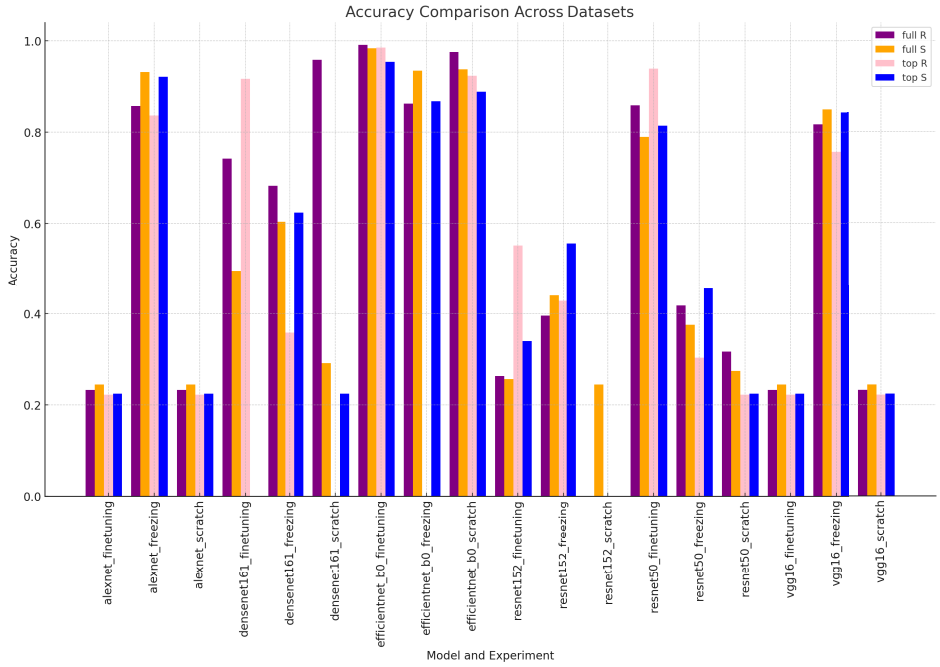


FIGURE 12. Accuracy across models and datasets.

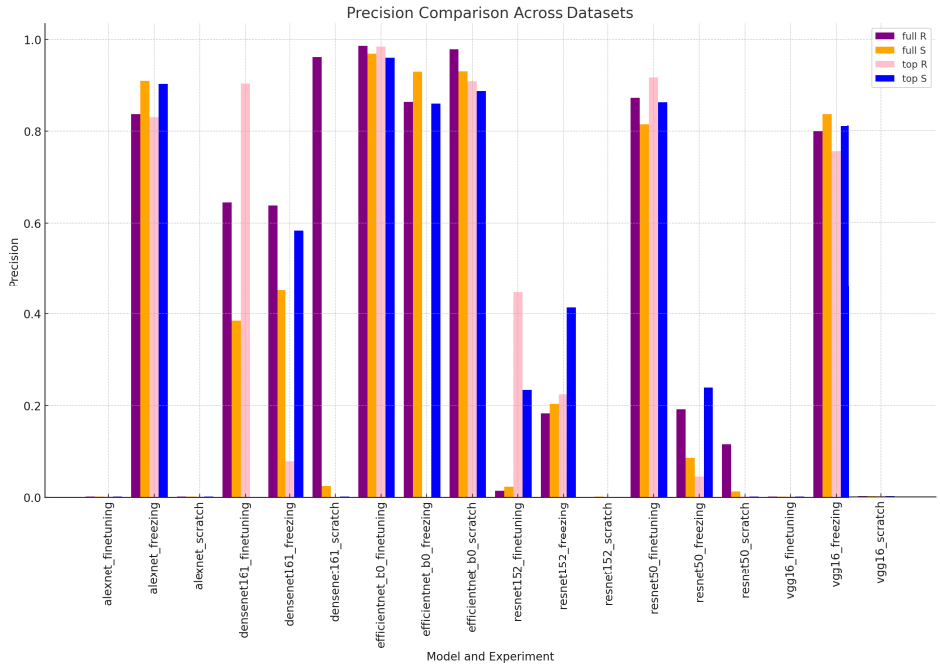


FIGURE 13. Precision across models and datasets.

0.9375, top R: 0.9239, top S: 0.8889) when trained without using pretrainied weights. In contrast, AlexNet, VGG16, ResNet152, and ResNet50 perform significantly worse. The convolutional architectures AlexNet and VGG16 reach accuracy values close to 0.23 in all scenarios. ResNet152 and ResNet50 perform the worst when training from scratch. Both architectures exhibit weak performance during the learning process. The best result among these two architectures is achieved by ResNet50, with an accuracy of 0.3167 on the dataset full R. ResNet152 demonstrates weaknesses, especially on full R, top R, and top S, where the architecture struggles to learn at all. This suggests that ResNet152 is not optimal for training from scratch.

Examining the precision in Table 4, EfficientNet-B0 dominates again with a value of 0.9782 on full R. AlexNet and VGG16 produce significantly worse results, with values close

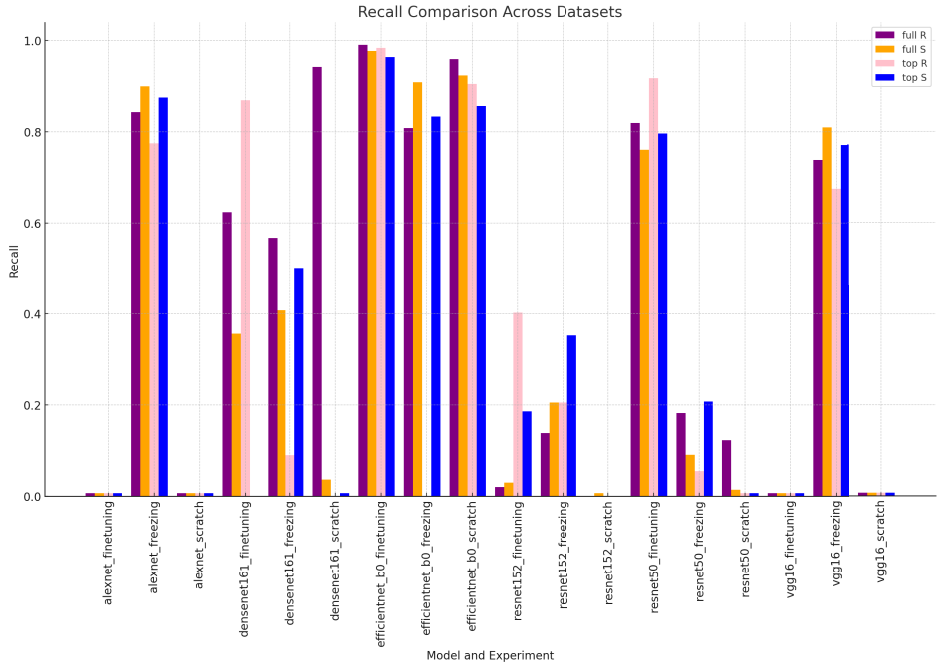


FIGURE 14. Recall across models and datasets.

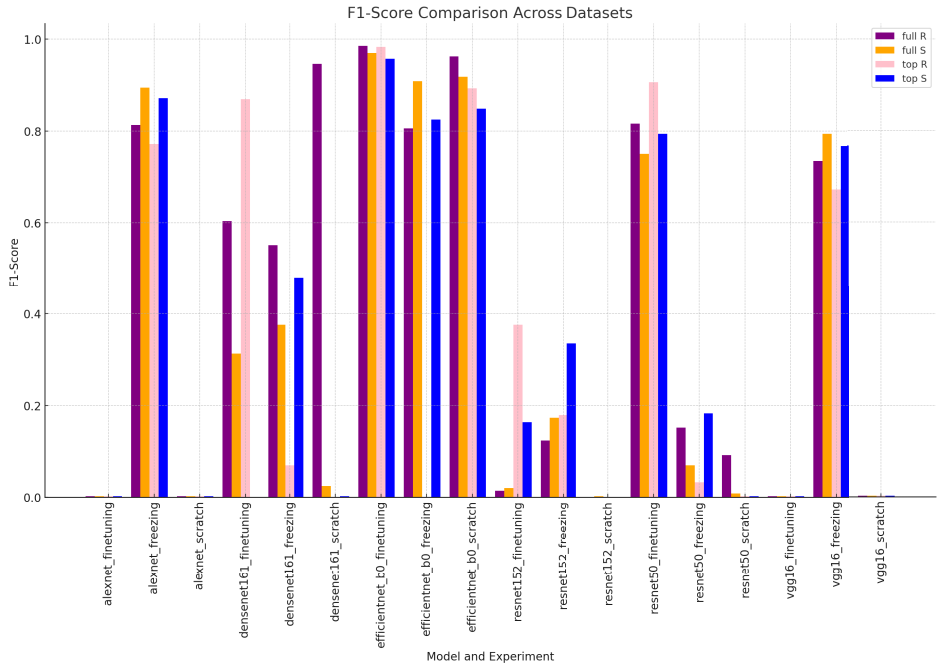


FIGURE 15. F1-Score across models and datasets.

to 0 (between 0.0015 and 0.0017). ResNet152 and ResNet50 also perform poorly:ResNet152 and ResNet50 also perform poorly: ResNet152 failed to train on full R and achieves only 0.0017 precision on full S. ResNet50 similarly exhibits weaknesses, with precision values such as 0.1159 (full R) and 0.0128 (full S).

The superiority of EfficientNet-B0 is also evident in recall (Table 5), particularly for full R (0.9592) and full S (0.9234). DenseNet161 follows closely behind on full R with 0.9421. However, DenseNet161 achieves only 0.037 on full S, performing even worse on top R and S datasets.

TABLE 7. Comparison of EfficientNetB0 performance on ToolSurface-144 dataset using freezing training strategy with literature from [34].

Dataset	Acc.	Prec.	Rec.	F1	Prec. Ref	Rec. Ref	F1 Ref	Acc. Ref	ΔPrec.	ΔRec.	ΔF1	ΔAcc.
Full R	0.8623	0.8638	0.8080	0.8055	0.9075	0.9100	0.9075	0.9061	0.0437	0.1020	0.1020	0.0438
Full S	0.9348	0.9296	0.9082	0.9082	0.9075	0.9100	0.9075	0.9061	0.0221	0.0018	0.0007	0.0287
Top R	0.9209	0.9087	0.9001	0.8927	0.9075	0.9100	0.9075	0.9061	0.0012	0.0099	0.0148	0.0148
Top S	0.8676	0.8598	0.8337	0.8246	0.9075	0.9100	0.9075	0.9061	0.0477	0.0763	0.0829	0.0385

The F1 score further confirms that EfficientNet-B0 performs best when training from scratch on all datasets, demonstrating its dominance over the other architectures.

B. TRAINING BY FINE-TUNING USING PRE-TRAINED MODEL

The ResNet50 architecture benefits the most from fine-tuning in a direct comparison. With fine-tuning, the accuracy for full R increases from 0.3167 to 0.8591, corresponding to an improvement of 171%. In other datasets, such as top S, the accuracy reaches 0.8143, a significant improvement over its inability to learn from scratch on this dataset, while the F1 score for full R even increases by 786%. These values clarify that ResNet50 relies heavily on pre-trained weights and hardly achieves meaningful results without them.

DenseNet161 also benefits from fine-tuning, but to a lesser extent than ResNet50. Accuracy on full S improves from 0.2917 (scratch) to 0.4959 (finetuning) (+70%), while the F1 score for full R changes from 0.9461 (scratch) to 0.6039 (finetuning), a decrease of 36.2%. Nevertheless, the model's performance remains inconsistent across all datasets, indicating that it does not generalize optimally in all scenarios. Even EfficientNet-B0, which already performs well when trained from scratch, still shows slight improvements through fine-tuning. For full R in particular, the accuracy increases from 0.9755 to 0.9910 (+1.6%), while the F1 score increases by 2.3%. This comparatively small improvement demonstrates that EfficientNet-B0 already achieves high performance without fine-tuning.

As seen in the results in Table 3-6, AlexNet and VGG16 do not benefit significantly from fine-tuning. Across all metrics, their values remain almost unchanged, and their precision and recall also remain extremely low, indicating that they are unsuitable for this task.

C. TRAINING BY FREEZING PARAMETERS USING PRE-TRAINED MODEL

A comparison of the third training method, parameter freezing, with the other two methods—training from scratch and fine-tuning—shows, based on the results from Tables 3–6, that parameter freezing leads to a significant performance increase for architectures like ResNet50, AlexNet, and VGG16 compared to training from scratch. However, its performance relative to full fine-tuning varies: for ResNet50 and EfficientNet-B0 it generally remains below that of full fine-tuning, whereas for AlexNet and VGG16, freezing achieves markedly better results.

Similar to fine-tuning, the ResNet50 architecture benefits particularly strongly from parameter freezing compared to training from scratch. While the model achieves an accuracy of only 0.3167 when trained from scratch on full R, freezing improves this value to 0.4179 (+32.0%). Nevertheless, it remains below the performance of full fine-tuning, which reaches an accuracy of 0.8591. This trend is also reflected in the F1 score for full R: ResNet50 reaches only 0.0921 during training from scratch but increases to 0.1520 (+65.0%) through freezing, which is substantially below the fine-tuning value of 0.8162 (+786% vs scratch). These results indicate that ResNet50 benefits greatly from pre-trained weights, especially when fully fine-tuned.

DenseNet161 shows varied results with parameter freezing. Accuracy on full S increases from 0.2917 (from scratch) to 0.6036 (freezing) (+106.9%) and surpasses the performance of full fine-tuning (0.4959) on this specific dataset. However, the F1 score for full R decreases from 0.9461 (scratch) to 0.5518 (freezing) (−41.7%). This demonstrates that although the model gains performance through freezing in some scenarios, its effectiveness is inconsistent across datasets and metrics.

In contrast to the other training methods for them, AlexNet and VGG16 benefit significantly from parameter freezing. For example, on the full R dataset, freezing boosts AlexNet's accuracy from 0.2328 (scratch) to 0.8574 and VGG16's accuracy from 0.2328 (scratch) to 0.8175. Similar large improvements are observed in precision, recall, and F1 score values (Tables 4-6), indicating that these models can leverage frozen pre-trained features effectively for this task, performing much better than when fine-tuned.

EfficientNet-B0 delivers excellent results across all three training methods. Comparing freezing to scratch on full R, Accuracy changes from 0.9755 (from scratch) to 0.8623 with freezing, while full fine-tuning increases it further to 0.9910. The F1 score on full R changes from 0.9625 (scratch) to 0.8055 (freezing). Despite the already high initial values when training from scratch, fine-tuning consistently provides the best performance for EfficientNet-B0, although freezing still yields strong results.

D. SUMMARIZING COMPARISON OF ALL THREE TRAINING METHODS

As can be seen in Figures 8 - 11, EfficientNet-B0 performs almost best in all metrics, Accuracy, Precision, Recall and F1-Score for all training strategies and datasets, except for

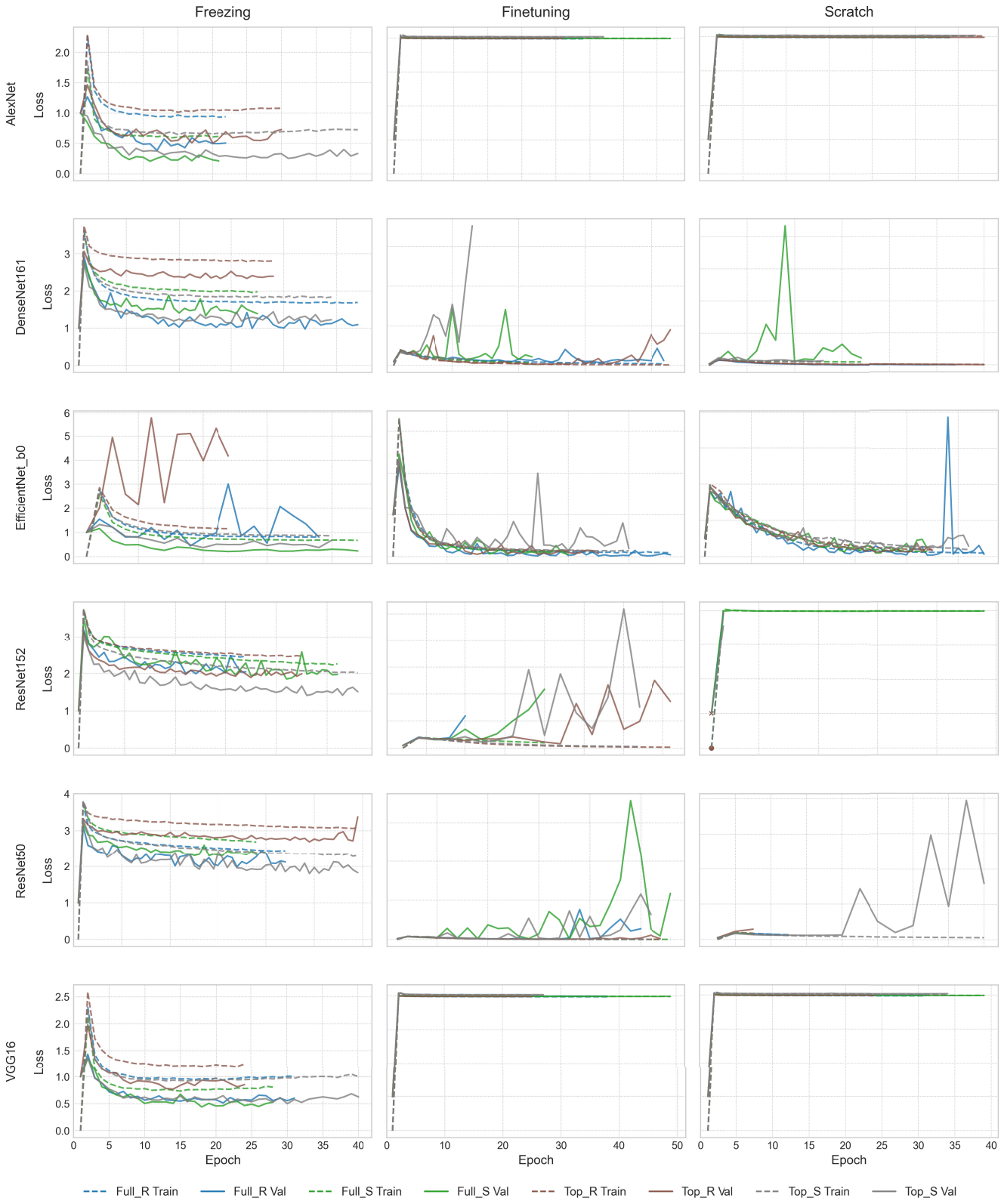


FIGURE 16. Loss curves of different architectures under various training strategies.

the Top R dataset and the freezing training strategy. The ResNet50 architecture benefits significantly from fine-tuning and parameter freezing and shows a significant increase in performance compared to training from scratch. ResNet152,

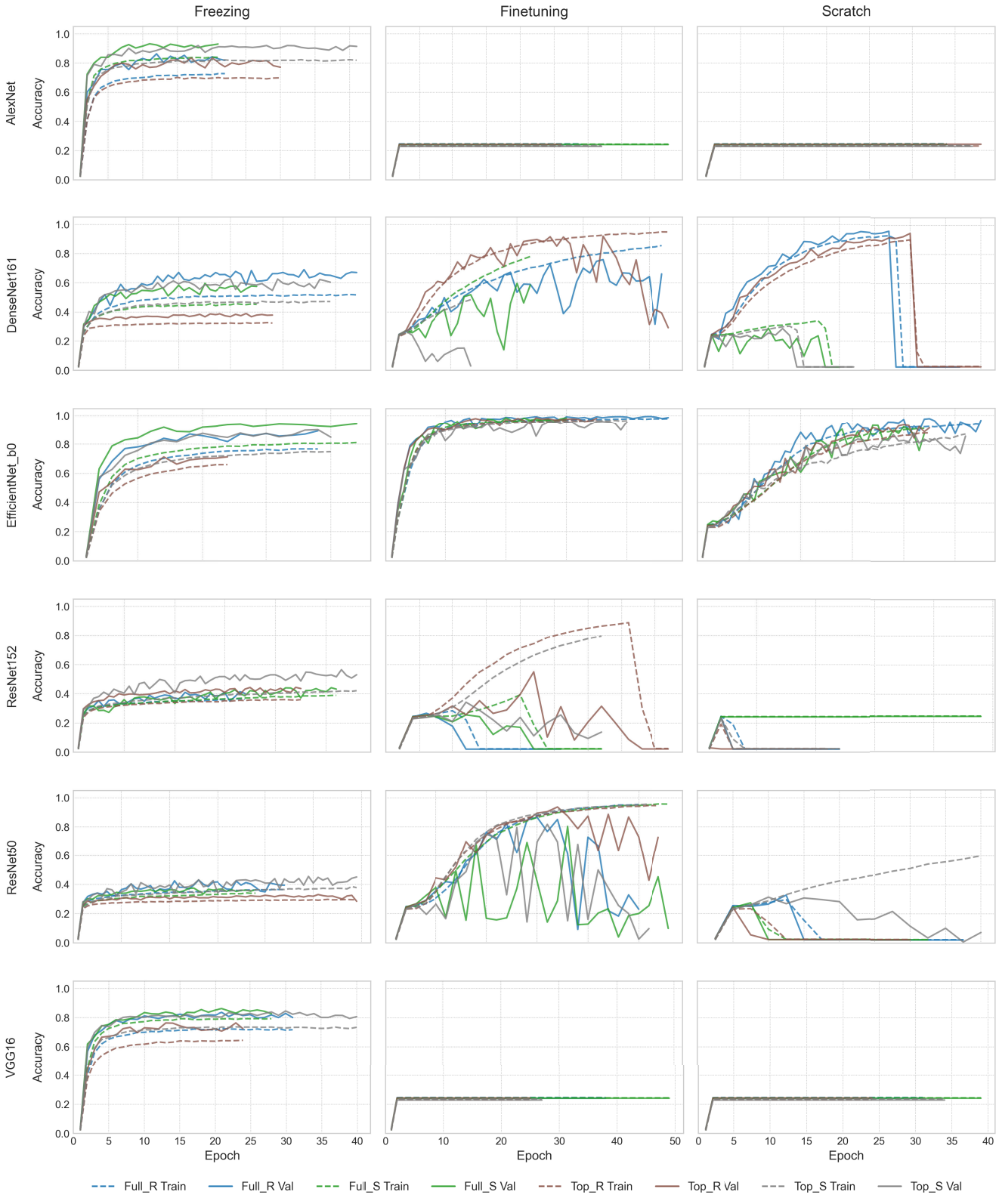


FIGURE 17. Accuracy curve during training.

AlexNet, and VGG16 consistently perform poorly in tool classification, exhibiting considerable difficulties in learning and generalization. DenseNet161 demonstrates mixed performance, making it only partially suitable for tool clas-

sification, with strong variations across different datasets. These results are consistent with the training behavior, i.e. the loss function, and the accuracy from Figures 16 and 17 in Appendix. It can be clearly seen that EfficientNet-B0 is able to train in almost all training strategies without exception and thus also achieves a stable accuracy value as the number of epochs increases. While other models such as ResNet-152 or ResNet-50 or VGG only deliver reasonably good results with the freezing training strategy. However, when it comes to fine-tuning or training from scratch, these models already fail during training in the first epochs.

E. EFFECTS OF LIGHTING CONDITIONS AND FIELD OF VIEW ON MODEL PERFORMANCE

Analysis of model performance using the metrics of accuracy, precision, recall and F1-score across models and datasets shows that illumination and image detail play a crucial role in classification performance in training neural networks (Figure 12,13,14,15). The Full R dataset (entire tool with diffuse ring light) provides the best results. A complete tool view with reflective features generates crucial information for reliable feature extraction. The Full S dataset (entire tool with homogeneous illumination) also achieves good results but performs slightly worse than Full R. This could be due to the reflection-reducing illumination highlighting fewer features on the cutting edges.

The Top R dataset (upper tool area with diffuse ring light) shows a significant reduction in performance compared to Full R. This could be because finer features, such as roughness in combination with reflections, are not reproducibly highlighted, whereas global features in Full R can be better utilized for feature extraction. Nevertheless, the performance in Top R remains higher than in Top S, suggesting that diffuse illumination is more beneficial for feature extraction despite the limited image detail. In contrast, Top S (upper tool area with homogeneous illumination) leads to an even greater reduction in performance, indicating that the lack of reflections and the higher depth structure of the details cause fluctuations in feature extraction.

The performance and sensitivity of the models across different datasets vary between architectures. EfficientNet-B0 remains relatively stable across all datasets and still achieves comparatively high accuracy even with Top R and Top S. ResNet50 and DenseNet161, on the other hand, perform well only with complete tool views (Full R and Full S) and are particularly susceptible to the limitations of smaller image sections. The fine-tuning and parameter freezing methods enable the models to achieve the best results, particularly for the complete tool views (Full R and Full S). Overall, Full R represents the best dataset for tool classification, while Top S performs the worst. This supports our assumption that cutting tools that are fully imaged and additionally show characteristic features due to the reflection patterns on the surfaces of the tools - especially on the cutting edges - are particularly highlighted by the reflections and are helpful for feature extraction. On the other hand, images in which only the upper area is taken into account also show finer structures such as surface roughness or subtle technical parameters such as chamfers for beveling edges to greater advantage, which means that the neural networks can be trained less effectively on global information.

In [34], the performance of different CNN architectures, including AlexNet, VGG16, ResNet50, and EfficientNetB0, was evaluated on medical datasets with tumor classes using transfer learning and the freezing strategy. Although our ToolSurface-144 dataset is much larger with 144 classes and the objects to be classified are more complex in many ways, our results are mostly consistent with those from [34], with the exception of ResNet50.

Particularly, the model EfficientNet-B0, which performs best, shows the following deviations on our datasets compared to the literature values from [34]:

Although the results are convincing - especially the performance of EfficientNet0 on almost all data sets with the exception of the ‘‘freezing’’ training strategy - there are some limitations, especially in an industrial environment. Tools differ not only in their geometry, but also in their coating. In addition, tools of the same type can differ both in size and in fine geometric details, which can ultimately lead to misclassifications. In the future, hybrid approaches to tool classification will be investigated, for example to reduce the dimensionality of the features and at the same time place tools in a global context.

V. CONCLUSION

In this work, a new dataset of optically critical tools, consisting of 144 different tool classes, was created and used to validate neural networks. A total of four different datasets were generated. Two datasets are based on conventional diffuse illumination, while the other two were acquired using an innovative and patented imaging technique. Two additional datasets, each containing only the upper section of the tool, were created to investigate the influence of global information, such as silhouettes, and local features, like phases, cutting edges, and other geometric details.

These four datasets were used to validate six state-of-the-art (SOTA) neural networks trained with three different methods: training from scratch, fine-tuning, and freezing of pre-trained weights. The neural networks’ validation showed that EfficientNet-B0 achieved the highest accuracy in almost all experiments and across all datasets. Especially in the case of fine-tuning, the model delivers nearly perfect results for tool classification. ResNet50 benefits greatly from fine-tuning and parameter freezing, showing a significant performance increase compared to training from scratch. ResNet152, AlexNet, and VGG16, on the other hand, show consistently poor classification performance and have significant difficulties in learning and generalization. DenseNet161

demonstrates mixed performance, making it only partially suitable for tool classification, with considerable variation across different datasets. The performance of the different models from different CNN architectures also varies depending on the trained dataset. EfficientNet-B0 remains relatively stable across all datasets and achieves comparatively high accuracy even with the poorly performing Top R and Top S datasets. ResNet50 and DenseNet161 achieve good results but only for complete tool views (Full R and Full S) and are particularly susceptible to the limitations of smaller image sections. The fine-tuning and parameter freezing methods yielded the best results, especially for complete tool views. Overall, Full R represents the best dataset for tool classification, while Top S delivers the worst results, presumably due to the lower level of detail in geometric and technical features.

A. FUTURE WORK

In future, other approaches will be considered to classify and test several tools simultaneously, instead of focusing on a single tool as in the present. An adaptation of the method from [54] - in particular the application of image subtraction and counting techniques - could be beneficial for tool classification and wear analysis. This assumes a mechanical setup that holds multiple tools in a rotationally symmetric centric and synchronized arrangement. Since the images in our system are already scaled to a uniform size, the framework from [54] could serve as a useful basis for developing a pipeline to classify multiple tools. In the future, however, other edge-device solutions will be considered where less computational power is required to classify tools. A promising approach would be a hybrid solution consisting of a Viola-Jones algorithm with Haar-like feature extraction and a cascade classifier [55]. Due to its computational efficiency compared to more complex deep learning frameworks such as ResNet or VGG-based models, this offers particular advantages for end-device applications and is particularly suitable for scenarios with limited computing resources, such as real-time inspection.

APPENDIX

TRAINING BEHAVIOR OF DIFFERENT ARCHITECTURES

See Figures 16 and 17.

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